



HONEY KAMLESH SONI

B.Tech. - Computer Science & Engineering

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PROFESSIONAL SUMMARY

Highly motivated and enthusiastic Computer Science student with strong interests in Machine Learning, Artificial Intelligence, Deep Learning, Web Development, and Networking. Experienced in leading technical and community-driven initiatives as the Chair of the IEEE MIT-ADT Student Branch WIE Affinity Group, where I successfully organised impactful events, coordinated teams, and promoted student engagement and women's empowerment in STEM. Passionate about innovation, problem-solving, and continuous learning, with strong leadership, communication, teamwork, and organisational skills, along with the ability to adapt quickly and work effectively in collaborative environments.

KEY EXPERTISE

Java Python MySQL Docker HTML CSS Machine Learning UI/UX MongoDB

EDUCATION

MIT School of Computing

2023 - 2027

B.Tech. - Computer Science & Engineering | CGPA: 9.29 / 10

The Orbis School, Keshavnagar, Pune

2023

12th | CBSE | Percentage: 85.60 / 100

Podar International School, Wagholi, Pune

2021

10th | CBSE | Percentage: 94.60 / 100

AWARDS AND SCHOLARSHIPS

- Secured 3rd Rank based on CGPA for outstanding academic performance
- Winner, MIT-ADT AI Grand Challenge 2026: Developed an AI-based weed detection and targeted spraying solution
- Best WIE Volunteer Award 2025

INTERNSHIPS

IEEE Computer Society Bangalore Chapter, Bangalore | Academics / Research

01 Apr, 2025 - 30 Sep, 2025

Research Intern

Key Skills: Analytics python data preprocessing Healthcare Machine Learning

Worked as a Research Intern focused on building a machine learning solution to predict cardiovascular disease risk, contributing to early detection in healthcare.

- Built and trained a machine learning model to predict the risk of cardiovascular disease and heart attacks
- Developed a data pipeline to clean, preprocess, and analyse large-scale patient health datasets
- Worked within a small agile team to define goals, manage tasks, and ensure the timely delivery of project milestones

IEEE EMBS Pune Chapter, India | Academics / Research

15 Jun, 2025 - 15 Jul, 2025

Student Intern

Key Skills:

Cognitive Wellness User Research UX Design React Native Gamification Mobile Application Development
UI Development

Worked as a Student Intern developing a mobile application aimed at improving cognitive health through gamified exercises.

- Created a React Native mobile app with gamified exercises to help users improve attention, memory, and cognitive performance
- Refined the app's interface and user flow to improve usability and overall user engagement

PROJECTS

Dhwani-Sanjeevani: AI-Based Heart Sound Analysis System

18 Feb, 2026 - 30 Apr, 2026

Mentor: Prof. Anant Kaulage | Team Size: 4

Key Skills:

- Python
- Deep Learning
- CNN
- Signal Processing
- Audio Processing
- Mel-Spectrogram Analysis
- Machine Learning

Dhwani-Sanjeevani is an AI-powered heart sound analysis system designed to detect cardiovascular abnormalities early, specifically for low-cost and accessible use in rural and resource-limited healthcare settings. The system operates by capturing heart sounds through a stethoscope and microphone module, processing the audio, and classifying it as normal or abnormal in real-time using a Convolutional Neural Network.

The audio signals were preprocessed using filtering, normalisation, and Mel-spectrogram conversion to extract clear features and improve prediction accuracy. Hardware components, including the ATmega2560 microcontroller, were integrated for real-time signal acquisition. The project covered end-to-end development, including model training, frontend interface design, real-time classification, and full hardware-software integration.

FasalAstra – AI-Powered Vision-Based Weed Detection with Targeted Spraying

25 Mar, 2026 - 17 Apr, 2026

Mentor: Dr. Rashmi S Nair | Team Size: 5

Key Skills:

- python
- computer vision
- yolov26n
- machine learning
- deep learning
- Real-Time Detection Systems
- Image Processing
- RaspberryPi 4B
- Pi Cam OV5647

FasalAstra is an autonomous, AI-powered precision agriculture system designed to optimise crop spraying by accurately distinguishing between crops and weeds in real time. The system utilises a custom-trained YOLO-based computer vision model with OpenCV running on a Raspberry Pi 4 for on-device processing and decision-making. Based on the detection results, a high-speed solenoid valve controlled through an IRFZ44N MOSFET enables precise spraying, while a mode-switching mechanism allows selective application of herbicides or fertiliser. This targeted approach reduces chemical wastage, lowers operational costs, and promotes sustainable farming practices.

MediVault: A Unified Health Records Platform

15 Aug, 2025 - 28 Nov, 2025

Mentor: Prof. Anant Kaulage | Team Size: 4

Key Skills:

- python
- html
- css
- javascript
- Frontend Web Development
- Authentication
- Authorization
- Data Security
- Data Privacy

MediVault is a healthcare web application built for secure digital storage and management of patient medical records, making health data easily accessible for both patients and healthcare providers. The application features a role-based authentication system to ensure data privacy, along with modules for record upload, retrieval, and organised data management.

Built using Python, the project involved implementing CRUD operations, handling sessions, and integrating the frontend and backend to ensure a smooth and secure data flow. Designed responsive user interfaces and optimised database queries for better performance and accessibility. Worked collaboratively within a team to develop, integrate, and test all core system functionalities.

Heart Attack Prediction using AIML

01 Apr, 2025 - 30 Sep, 2025

Mentor: Prof. Monika Dongare | Team Size: 4

- Key Skills:
- AI
- Medical
- Heart Attack Prediction
- Machine Learning
- Python
- Random Forest

Built an AI-powered heart attack prediction system using machine learning to analyse key medical parameters, including age, cholesterol, blood pressure, heart rate, and ECG data, designed to support early clinical decision-making and preventive healthcare. Developed and evaluated three models, including Random Forest, Logistic Regression, and Support Vector Machine, using Python, Scikit-learn, Pandas, and NumPy, with thorough feature engineering and preprocessing techniques such as missing value handling, feature selection, normalisation, and stratified cross-validation to improve reliability and prevent overfitting. The Random Forest model achieved approximately 98% prediction accuracy during testing across multiple datasets. Also built a lightweight web interface using HTML and CSS and handled complete frontend-backend integration to enable real-time prediction generation from user inputs.

NeuroPlay: Sharpen & Focus

15 Jun, 2025 - 15 Jul, 2025

Mentor: Prof. Trupti Kudale | Team Size: 2

Key Skills:

- React Native
- Javascript
- UI Development
- UX Design
- User Research
- Mobile Application Development
- Gamification

NeuroPlay is a cross-platform mobile application built to help users sharpen cognitive skills through interactive brain-training exercises, covering areas like memory, attention, focus, and reaction time. Developed using React Native and JavaScript, the app delivers a smooth and engaging experience across both Android and iOS platforms.

Worked across the full frontend and application layer, building responsive interfaces and integrating game logic across multiple modules to ensure each activity felt interactive and fluid. Implemented core features, including score tracking, level progression, timer-based activities, and performance monitoring to keep users engaged and aware of their progress. Focused on optimising app performance and navigation flow to minimise lag and deliver a seamless experience, while also ensuring the UI remained consistent, accessible, and visually clean across different screen sizes and devices.

Mentor: Prof. Trupti Kudale | Team Size: 4

Key Skills: HTML CSS Javascript PHP MySQL

AgriMart is a full-stack e-commerce platform built to directly connect local farmers with consumers and retailers, creating a fair-trade marketplace by cutting out intermediaries and improving market accessibility for agricultural communities. The platform allows farmers to list and showcase their products while buyers can browse and place orders through a simple, intuitive interface. Built using HTML, CSS, JavaScript, and PHP, the application covers the full development stack from database design and backend logic to frontend implementation. Worked on building and connecting the MySQL database to handle user accounts, product catalogues, and order management, ensuring reliable and efficient data flow across the system. Developed and integrated backend modules with frontend components, making sure every user interaction from browsing products to placing orders worked seamlessly end-to-end. Also focused on responsive UI design across both farmer and buyer modules to deliver a clean and accessible experience across devices.

ACHIEVEMENTS

- Winner of the MIT ADT AI Grand Challenge 2026 for developing an AI-based agriculture solution on vision-based weed detectionwith targeted spraying, and awarded a cash prize of Rs. 1,00,000 for innovation and technical implementation.

ASSESSMENTS / CERTIFICATIONS

Neural Networks and Deep Learning

Provider: DeepLearning.AI, Coursera

Aggregate: 98.42 / 100

Key Skills:

Deep Learning AI Model Training Python CNN Supervised Learning Model Optimization RNN ANN

Advanced Java

Provider: LearnQuest, Coursera

Aggregate: 96.70 / 100

Key Skills: Spring Boot Maven Cloud Development Restful API Java Scalability

Practical Machine Learning

Provider: Johns Hopkins University, Coursera

Aggregate: 93 / 100

Key Skills: Machine Learning Data Preprocessing Supervised Learning Feature Engineering Model Training

Programming Foundations with JavaScript, HTML and CSS

Provider: Duke University, Coursera

Aggregate: 87.42 / 100

Key Skills: HTML CSS Javascript Computational Thinking Web Design Web Development UI Development

Docker for Beginners with Hands-on labs

Provider: KodeKloud, Coursera

Aggregate: 85.75 / 100

Key Skills: Docker Linux Commands Computer Networking CLI YAML Containerization

POSITIONS OF RESPONSIBILITY

Chair, IEEE MIT-ADT Student Branch Women in Engineering Affinity Group

Served as the Chair of the IEEE WIE MIT-ADT Student Branch, passionately leading a dynamic team of 25 members and driving impactful technical, professional, and women empowerment initiatives. Played a key role in planning and executing engaging events while fostering leadership, collaboration, and active student involvement across the IEEE community. Successfully organised inspiring events such as the Women’s Day Debate and Thought-Provoking Challenge, self-defence awareness sessions with Mission Sahasi, interactive IEEE membership drives involving 50+ students and many more. Enthusiastically worked towards empowering women in STEM, encouraging innovation, and creating a vibrant and inclusive student community.

CO-CURRICULAR ACTIVITIES

- Active member of IEEE MIT-ADT Student Branch and Chair, IEEE MIT-ADT Student Branch WIE Affinity Group
- Active participation in Hackathons to gain confidence, skills and experience

EXTRA CURRICULAR ACTIVITIES

- o Participated in the Swami Vivekanand Marathon 2026 and Lokmat Maha Marathon 2026, Pune

PERSONAL INTERESTS / HOBBIES

- o Playing badminton
- o Swimming
- o Learning new skills

WEB LINKS

- o Github - <https://github.com/Honey-Soni-05>

PERSONAL DETAILS

Gender: Female

Marital Status: Single

Current Address: Gera World of Joy, Ubale Nagar, Wagholi,
Pune, Maharashtra, India - 412207

Email: honeysoni321@gmail.com

Date of Birth: 21 Nov, 2005

Known Languages: English, Hindi, Gujarati, German

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